**Course Name:** Basic Electrical Engineering
**Course Code:** EE-101
**Course Type:** Core

**Contact Hours/Week:** 2L **Course Credits:** 02

**Course Objectives**

- To impart knowledge about the electrical quantities and to understand the impact of electricity in a global and societal context.
- To introduce the fundamental concepts relevant to DC and AC circuits and network theorems.
- Highlight the importance of electromagnetism and transformers in transmission and distribution of electric power.
- To explain the working principle, construction, applications of DC machines, AC machines & measuring instruments.

| Unit Number | Course Content | Contact Hours |
|---|---|---|
| UNIT-01 | **Introduction:** Circuit elements–RLC, voltage and current sources, Fundamental Laws of Electrical Engineering, Methods of Circuits analysis, Network theorems, Generation of A.C. sinusoidal voltage and currents, average and r.m.s. values, Form factor and peak factor, phasor representation. | 07L |
| UNIT-02 | **AC Circuits,Domestic Electric Wiring & Storage Batteries:** Analysis of single phase series, parallel and series-parallel circuits, Active and reactive power, p.f. and volt-amperes, frequency response and Q-factor, Analysis of balanced and unbalanced three phase a.c. circuits – Concept of voltage, current and power in three phase circuits, Basics of Domestic Electric Wiring and Storage Batteries. | 06L |
| UNIT-03 | **Magnetic Circuits and Electrical machines:** Magnetic circuit concept, B-H curves characteristics of magnetic materials, practical magnetic circuits, magnetic circuits with D.C. and A.C. excitation, hysteresis and eddy current losses, Magnetic force, self and mutual inductances, Principle of Transformer operation, construction and equivalent circuit of transformer, Phasor diagram and tests, Fundamentals of D.C. and A.C. machines.Principles of Power generation and Industrial Power Distribution. | 08L |
| UNIT-04 | **Measuring Instruments:** Classification of Instruments, Introduction to galvanometer (Moving coil and moving iron), ammeter, voltmeter, wattmeter, use of shunt and multiplier. | 03L |

**Course Outcomes**

Upon successful completion of the course, the students will be able to

CO1: Identify and predict the behavior of any electrical and magnetic circuit.
CO2: Formulate and solve complex AC and DC circuits.
CO3: Realize the requirement of transformers in transmission and distribution of electric power and other applications.
CO4: Identify the type of electrical machines used for that particular application.

**Textbooks**


1. Fundamental of Electric Circuits by Charles K Alexander and Matthew N. O. Sadiku, TMH Publication.
2. Electrical Engineering Fundamentals by Vincent Del Toro, PHI Publication.
3. Basic Electrical Engineering by V N Mittal & Arvind Mittal, TMH Publication.
4. Basic Electrical Engineering by T.K. Nagsarkar & M.S. Sukhija, Oxford University Press.


**Reference Books**


1. Basic Electrical Technology by A.E. Fitzgerald, McGraw Hill Publication.
2. Electrical Estimating and Costing by N Alagappan and B Ekambaram, TMH Publication.
3. Electrical Technology by H. Cotton, CBS Publishers & Distributors.